

BAYESIAN INFERENCE ON SHIFTED LINDLEY DISTRIBUTION BASED ON DIFFERENT LOSS FUNCTIONS



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- Introduction:

- Bayesian inference has emerged as a powerful tool in statistical modeling, offering a flexible framework for incorporating prior knowledge into the analysis of data.
- Bayesian inference on the shifted Lindley distribution involves using observed data and prior beliefs to update our knowledge about the parameters of this distribution.
- The process incorporates various loss functions to evaluate the accuracy and cost of our estimations.
- Bayesian inference combines prior information about a parameter with observed data to form a posterior distribution.
- This project focuses on Bayesian inference techniques applied to the shifted Lindley distribution, exploring the impact of different loss functions on parameter estimation.

Shifted Lindley Distribution:

- The probability density function (pdf) of the Shifted Lindley (SL(μ , θ)) distribution from Maiti et al. (2021).

$$f(x; \theta, \mu) = \frac{\theta^2}{1 + \theta(1 + \mu)} (1 + x) e^{-\theta(x - \mu)}, x > \mu > 0$$

- The CDF of a Shifted Lindley distribution with parameters (θ , μ) is given by

$$F(x; \theta, \mu) = 1 - \frac{1 + \theta(1 + x)}{1 + \theta(1 + \mu)} e^{-\theta(x - \mu)}, x > \mu > 0$$

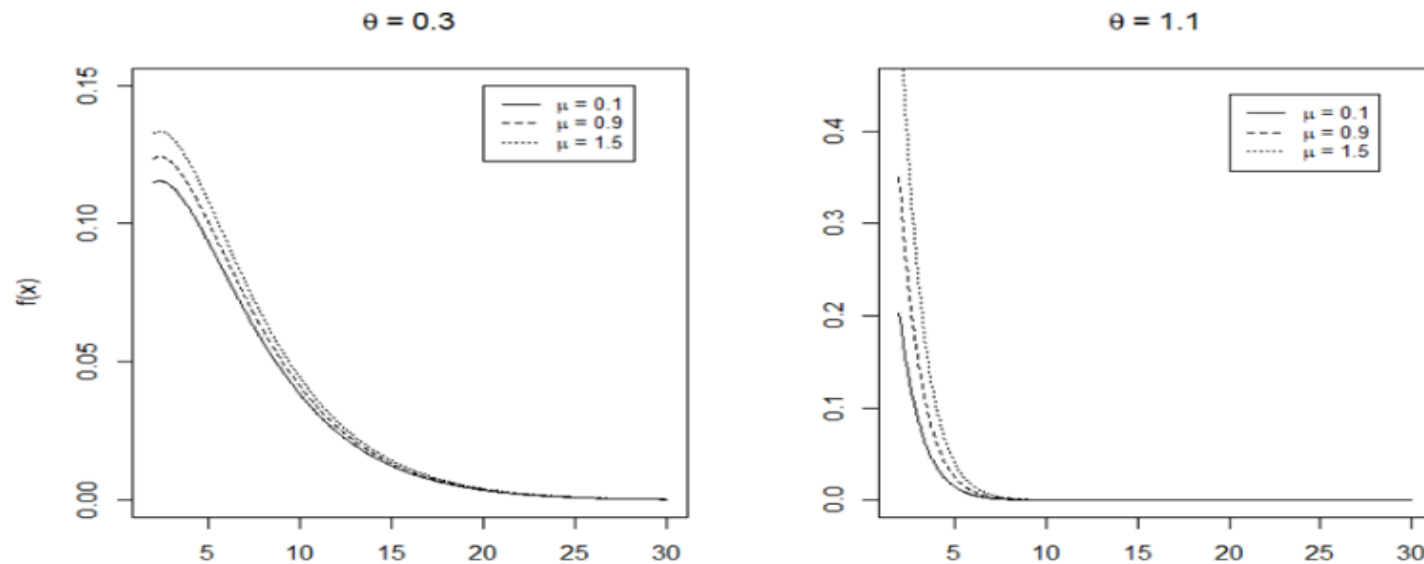


Figure 1: The PDF's of various Shifted Lindley distribution for different values of parameters.

The shape of the shifted Lindley distribution depends on its parameters μ and θ . Figure 1 illustrates the pdf and cdf for various values of these parameters.

- Smaller θ : The pdf is right-skewed, indicating a longer tail on the right side.
- Larger θ : The pdf resembles an inverted J shape, tapering off to look like a standard exponential curve, indicating a rapid decline in probability for larger values.

❖ Some properties of $SL(\mu, \theta)$:

1. Mean:

Mean of the shifted Lindley distribution is

$$\mu'_1 = E(X) = \frac{\theta^2}{1 + \theta(1 + \mu)} \int_{\mu}^{\infty} x(1 + x)e^{-\theta(x-\mu)} dx = \mu + \frac{2}{\theta} - \frac{1 + \mu}{1 + \theta(1 + \mu)}$$

2. Variance:

The variance of the $SL(\mu, \theta)$ is

$$\mu_2 = \mu'_2 - \mu_1'^2 = \frac{2}{\theta^2} - \frac{(1 + \mu)^2}{(1 + \theta(1 + \mu))^2}.$$

3. Skewness:

$$\mu_3 = \frac{4}{\theta^3} - \frac{2(1 + \mu)^3}{(1 + \theta(1 + \mu))^3}.$$

The skewness of the shifted Lindley is calculated from this equation

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3},$$

4. Kurtosis:

$$\mu_4 = \frac{24}{\theta^4} - \frac{3(1 + \mu)^4}{(1 + \theta(1 + \mu))^4} - \frac{12(1 + \mu)^2}{\theta^2(1 + \theta(1 + \mu))^2}.$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2}.$$

5. Reliability or survival function:

The survival function of the SL(μ , θ) distribution is

$$R(t; \mu, \theta) = 1 - F(t; \mu, \theta)$$

$$= 1 - \left(1 - \frac{1 + \theta(1 + t)}{1 + \theta(1 + \mu)} e^{-\theta(t - \mu)}\right) = \frac{1 + \theta(1 + t)}{1 + \theta(1 + \mu)} e^{-\theta(t - \mu)}.$$



Loss Function:

A loss function is a crucial concept that measures the cost associated with the difference between the estimated value and the true value of a parameter. The goal of Bayesian estimation is to minimize this loss, leading to more accurate and reliable estimates.

Different loss functions:

In this project, I will use three types of loss function such as, squared error(SE) loss function, linex (LE) loss function, general entropy(GE) loss function.

1. squared error(SE) loss function: A very popular quadratic loss is the squared error(SE) loss function given by
Where $\hat{g}(\theta)$ is an estimate of the parametric function $g(\theta)$.

$$\Delta_{SE}(g(\theta), \hat{g}(\theta)) = (g(\theta) - \hat{g}(\theta))^2.$$

The Bayes estimate of $g(\theta)$ against SE loss function is the posterior mean is given by $\hat{g}_{SE}(\theta) = E_{\theta}[g(\theta) \mid \text{data}]$.

2. Linex (LE) loss function: The Linex loss function is defined as $\Delta_{LE}(g(\theta), \hat{g}(\theta)) = \exp(\nu(g(\theta) - \hat{g}(\theta))) - \nu(g(\theta) - \hat{g}(\theta)) - 1, \quad \nu \neq 0,$

The Bayesian estimate of $g(\theta)$ based on the Linex loss function is given by $\hat{g}_{LE}(\theta) = -\frac{1}{\nu} \log E_{\theta}[\exp(-\nu g(\theta)) \mid \text{data}]$.

3. General entropy(GE) loss function: The General Entropy (GE) loss function is defined as, $\Delta_{GE}(g(\theta), \hat{g}(\theta)) \propto \left(\frac{\hat{g}(\theta)}{g(\theta)}\right)^{\omega} - \omega \log\left(\frac{\hat{g}(\theta)}{g(\theta)}\right) - 1, \quad \omega \neq 0$
The Bayesian estimate of $g(\theta)$ based on the GE loss function is given by

$$\hat{g}_{GE}(\theta) = (E_{\theta}[(g(\theta))^{-\omega} \mid \text{data}])^{-1/\omega}.$$

Bayesian analyses:

The likelihood function is given by
$$l(X, \mu, \theta) = \frac{\theta^{2n}}{(1 + \theta(1 + \mu))^n} \prod_{i=1}^n (1 + x_i) e^{-\theta \sum_{i=1}^n (x_i - \mu)}$$

The prior distributions on the parameters are
$$\pi_1(\theta; a_1, b_1) \propto \theta^{a_1-1} e^{-b_1 \theta}, \quad \theta > 0$$
$$\pi_2(\mu; a_2, b_2) \propto \mu^{a_2-1} e^{-b_2 \mu}, \quad \mu > 0$$

Where, the hyperparameters a_i and b_i , $i=1,2$

We know that, $\text{Posterior}(\theta, \mu | \text{data}) \propto \text{likelihood}(\text{data} | \theta, \mu) \times \text{prior}(\theta) \times \text{prior}(\mu)$

Now, the posterior joint probability density function of θ and μ given that

$$\pi^*(\theta, \mu | x) = \frac{\theta^{2n+a_1-1}}{C(1 + \theta(1 + \mu))^n} e^{-\theta(b_1 + \sum_{i=1}^n (x_i - \mu))} e^{-b_2 \mu} \mu^{a_2-1} \left[\prod_{i=1}^n (1 + x_i) \right].$$

Where,

$$C = \int_0^\infty \int_0^\infty \frac{\theta^{2n+a_1-1}}{(1 + \theta(1 + \mu))^n} e^{-\theta(b_1 + \sum_{i=1}^n (x_i - \mu))} e^{-b_2 \mu} \mu^{a_2-1} \left[\prod_{i=1}^n (1 + x_i) \right] d\theta d\mu.$$

❖ Now, we derive the Bayes estimates of θ , μ and the reliability parameter $R(t)$.

❖ Under squared error(SE) loss function:

$$\hat{\theta}_{SE} = E[\theta|X] = \frac{1}{c} \int_0^\infty \int_0^\infty \frac{\theta^{2n+a_1}}{(1+\theta(1+\mu))^n} e^{-\theta(b_1+\sum_{i=1}^n(x_i-\mu))} e^{-b_2\mu} \mu^{a_2-1} \left[\prod_{i=1}^n (1+x_i) \right] d\theta d\mu,$$

$$\hat{\mu}_{SE} = E[\mu|X] = \frac{1}{c} \int_0^\infty \int_0^\infty \frac{\theta^{2n+a_1-1}}{(1+\theta(1+\mu))^n} e^{-\theta(b_1+\sum_{i=1}^n(x_i-\mu))} e^{-b_2\mu} \mu^{a_2} \left[\prod_{i=1}^n (1+x_i) \right] d\theta d\mu$$

The Bayes estimate of $R(t)$ against the squared error loss function can be derived

$$\hat{R}_{SE}(t) = E\left[\frac{1+\theta(1+t)}{1+\theta(1+\mu)} e^{-\theta(t-\mu)} | X\right] = \frac{1}{c} \int_0^\infty \int_0^\infty \frac{\theta^{2n+a_1-1}}{(1+\theta(1+\mu))^n} e^{-\theta(b_1+\sum_{i=1}^n(x_i-\mu))} e^{-b_2\mu} \mu^{a_2-1} \left[\prod_{i=1}^n (1+x_i) \right] \frac{1+\theta(1+t)}{1+\theta(1+\mu)} e^{-\theta(t-\mu)} d\theta d\mu$$

❖ Under linex(LE) loss function:

$$\hat{\theta}_{LE} = -\frac{1}{\nu} \log(E[\exp(-\nu\theta) | X]) = -\frac{1}{\nu} \log\left[\frac{1}{c} \int_0^\infty \int_0^\infty \frac{\theta^{2n+a_1-1}}{(1+\theta(1+\mu))^n} e^{-\theta(b_1+\sum_{i=1}^n(x_i-\mu)+\nu)} e^{-b_2\mu} \mu^{a_2-1} \left[\prod_{i=1}^n (1+x_i) \right] d\theta d\mu\right]$$

$$\hat{\mu}_{LE} = -\frac{1}{\nu} \log(E[\exp(-\nu\mu) | X]) = -\frac{1}{\nu} \log\left[\frac{1}{c} \int_0^\infty \int_0^\infty \frac{\theta^{2n+a_1-1}}{(1+\theta(1+\mu))^n} e^{-\theta(b_1+\sum_{i=1}^n(x_i-\mu))} e^{-\mu(b_2+\nu)} \mu^{a_2-1} \left[\prod_{i=1}^n (1+x_i) \right] d\theta d\mu\right]$$

The Bayes estimate of $R(t)$ against the linex loss function can be derived

$$\hat{R}(t)_{LE} = -\frac{1}{\nu} \log \mathbb{E}[\exp(-\nu R(t)) | x] = -\frac{1}{\nu} \log \frac{1}{c} \int_0^\infty \int_0^\infty \frac{\theta^{2n+a_1-1}}{(1+\theta(1+\mu))^n} e^{-\theta(b_1+\sum_{i=1}^n(x_i-\mu))} e^{-b_2\mu} \mu^{a_2-1} \left[\prod_{i=1}^n (1+x_i) \right] \exp\left[-\nu \frac{1+\theta(1+t)}{1+\theta(1+\mu)} e^{-\theta(t-\mu)}\right] d\theta d\mu$$

❖ Under general entropy(GE) loss function:

$$\hat{\theta}_{GE} = \mathbb{E}[\theta^{-w} | X]^{-\frac{1}{w}}$$

The conditional expectation can be calculated as-

$$\mathbb{E}[\theta^{-w} | X] = \frac{1}{c} \int_0^\infty \int_0^\infty \frac{\theta^{2n+a_1-1-w}}{(1+\theta(1+\mu))^n} e^{-\theta(b_1+\sum_{i=1}^n(x_i-\mu))} e^{-b_2\mu} \mu^{a_2-1} \left[\prod_{i=1}^n (1+x_i) \right] d\theta d\mu$$

and

$$\hat{\mu}_{GE} = \mathbb{E}[\mu^{-w} | X]^{-\frac{1}{w}}$$

The conditional expectation can be calculated as-

$$\mathbb{E}[\mu^{-w} | X] = \frac{1}{c} \int_0^\infty \int_0^\infty \frac{\theta^{2n+a_1-1}}{(1+\theta(1+\mu))^n} e^{-\theta(b_1+\sum_{i=1}^n(x_i-\mu))} e^{-b_2\mu} \mu^{a_2-1-w} \left[\prod_{i=1}^n (1+x_i) \right] d\theta d\mu$$

The Bayes estimate of R(t) against the general entropy loss function can be derived

$$\hat{R}_{GE}(t) = E\left[\left(\frac{1+\theta(1+t)}{1+\theta(1+\mu)}\right)^{-w} e^{\theta w(t-\mu)} | X\right]^{-\frac{1}{w}}$$

then

$$E\left[\left(\frac{1+\theta(1+t)}{1+\theta(1+\mu)}\right)^{-w} e^{\theta w(t-\mu)} | X\right] = \frac{1}{c} \int_0^\infty \int_0^\infty \frac{\theta^{2n+a_1-1}}{(1+\theta(1+\mu))^n} e^{-\theta(b_1+\sum_{i=1}^n(x_i-\mu))} e^{-b_2\mu} \mu^{a_2-1} \left[\prod_{i=1}^n (1+x_i) \right] \left(\frac{1+\theta(1+t)}{1+\theta(1+\mu)}\right)^{-w} e^{w\theta(t-\mu)} d\theta d\mu$$

❑ Computation challenge:

The above bayes estimators are involved the double integrals for which simple closed forms cannot be obtained.so, we turn to two popular approximation techniques I.e, Lindley approximation and Markov chain Monte carlo(MCMC) in the following sections to derive approximate bayes estimate for the parameters.

❖ Lindley tries to approximate the posterior function itself, whereas MCMC tries to approximate the distribution by simulating from simpler distributions.

Lindley Approximation:

We use Lindley technique for computing the bayes estimates . Let $h(\theta, \mu)$ be any function of the parameters

$$\mathbb{E}[h(\theta, \mu) | x] = \int_0^\infty \int_0^\infty h(\theta, \mu) \pi^*(\theta, \mu | x) d\theta d\mu = \frac{\int_0^\infty \int_0^\infty h(\theta, \mu) \exp(L(\theta, \mu; x) + \eta(\theta, \mu)) d\theta d\mu}{\int_0^\infty \int_0^\infty \exp(L(\theta, \mu; x) + \eta(\theta, \mu)) d\theta d\mu}$$

Where, the function $\eta(\theta, \mu) = \log \pi_1(\theta; a_1, b_1) + \log \pi_2(\mu; a_2, b_2)$, and $L(\theta, \mu; x)$ is the loglikelihood function of the parameters θ and μ given by

$$L(X, \mu, \theta) = \log l(X, \mu, \theta) = \log\left(\frac{\theta^{2n}}{(1 + \theta(1 + \mu))^n} \prod_{i=1}^n (1 + x_i) e^{-\theta \sum_{i=1}^n (x_i - \mu)}\right) = 2n \log \theta - n \log(1 + \theta(1 + \mu)) + \sum_{i=1}^n \log(1 + X_i) - \theta \sum_{i=1}^n (X_i - \mu)$$

The Bayes estimate of $h(\theta, \mu)$ is approximated by

$$E[h(\theta, \mu) | x] \approx h(\theta, \mu) + \frac{1}{2} \sum_{i=1}^2 \sum_{j=1}^2 h_{ij} \tau_{ij} + \sum_{i=1}^2 \eta_i V_i + \frac{1}{2} \sum_{i=1}^2 L_{iii} \tau_{ii} V_i + \frac{1}{2} (L_{112} (2\tau_{12} V_1 + \tau_{11} V_2) + L_{122} (\tau_{22} V_1 + 2\tau_{12} V_2))$$

Where $h_1 = \frac{\partial h}{\partial \theta}, h_2 = \frac{\partial h}{\partial \mu}, h_{11} = \frac{\partial^2 h}{\partial \theta^2}, h_{22} = \frac{\partial^2 h}{\partial \mu^2}, h_{12} = h_{21} = \frac{\partial^2 h}{\partial \theta \partial \mu}$, and $V_r = \sum_j h_j \tau_{rj}$

with τ_{ij} being (i,j)th elements of the inverse of the matrix $[-\frac{\partial^2 L}{\partial \theta \partial \mu}]$

Calculating all the expression in above expectation at the maximum likelihood estimators of the parameters θ and μ .

❖ The required quantities in the expression $E[h(\theta, \mu)|x]$ are derived as

$$L_1 = \left. \frac{\partial L(\theta, \mu)}{\partial \theta} \right|_{(\hat{\theta}, \hat{\mu})} = \frac{2n}{\theta} - \frac{n(1 + \mu)}{1 + \theta(1 + \mu)} - \sum_{i=1}^n (x_i - \mu)$$

$$L_{11} = \left. \frac{\partial^2 L(\theta, \mu)}{\partial \theta^2} \right|_{(\hat{\theta}, \hat{\mu})} = -\frac{2n}{\theta^2} + \frac{n(1 + \mu)^2}{(1 + \theta(1 + \mu))^2}$$

$$L_{111} = \left. \frac{\partial^3 L(\theta, \mu)}{\partial \theta^3} \right|_{\hat{\theta}=\hat{\theta}, \mu=\hat{\mu}} = \frac{4n}{\theta^3} - \frac{2n(1 + \mu)^3}{(1 + \theta(1 + \mu))^3}$$

$$L_2 = \left. \frac{\partial L(\theta, \mu)}{\partial \mu} \right|_{(\hat{\theta}, \hat{\mu})} = \frac{-n\theta}{1 + \theta(1 + \mu)} + n\theta$$

$$L_{22} = \left. \frac{\partial^2 L(\theta, \mu)}{\partial \mu^2} \right|_{\hat{\theta}=\hat{\theta}, \mu=\hat{\mu}} = \frac{n\theta^2}{(1 + \theta(1 + \mu))^2}$$

$$L_{222} = \left. \frac{\partial^3 L(\theta, \mu)}{\partial \mu^3} \right|_{\hat{\theta}=\hat{\theta}, \mu=\hat{\mu}} = \frac{-2n\theta^3}{(1 + \theta(1 + \mu))^3}$$

$$L_{12} = L_{21} = \left. \frac{\partial^2 L(\theta, \mu)}{\partial \theta \partial \mu} \right|_{(\hat{\theta}, \hat{\mu})} = \frac{-n}{(1 + \theta(1 + \mu))^2} + n$$

$$L_{112} = \left. \frac{\partial^3 L(\theta, \mu)}{\partial \theta^2 \partial \mu} \right|_{\hat{\theta}=\hat{\theta}, \mu=\hat{\mu}} = \frac{(1 + \theta(1 + \mu))^2 2n(1 + \mu) - n(1 + \mu)^2 2\theta(1 + \theta(1 + \mu))}{(1 + \theta(1 + \mu))^4}$$

$$L_{122} = \left. \frac{\partial^3 L(\theta, \mu)}{\partial \theta \partial \mu^2} \right|_{\hat{\theta}=\hat{\theta}, \mu=\hat{\mu}} = \frac{(1 + \theta(1 + \mu))^2 2n\theta - 2n\theta^2(1 + \mu)(1 + \theta(1 + \mu))}{(1 + \theta(1 + \mu))^4}$$

$$\eta_1 = \frac{a_1 - 1}{\hat{\theta}} - b_1,$$

$$\eta_2 = \frac{a_2 - 1}{\hat{\mu}} - b_2$$

Then, we derive the approximate Bayes estimate of θ , μ and $R(t)$ under the considered loss functions.

❖ Bayes Estimates Using Squared Error(SE) Loss Function:

- For computing the estimate of θ under SE loss function. Let $h(\theta, \mu) = \theta$. Hence, $h_1=1, h_2=h_{11}=h_{21}=h_{22}=h_{21}=0$ and we have

$$\hat{\theta}_{SE} \approx \hat{\theta} + \tau_{11}\eta_1 + \tau_{21}\eta_2 + \frac{1}{2} (L_{111}\tau_{11}^2 + L_{222}\tau_{22}\tau_{12} + 3L_{112}\tau_{12}\tau_{11})$$

- Similarly, setting $h(\theta, \mu) = \mu$, we have $h_2 = 1, h_1 = h_{11} = h_{21} = h_{12} = h_{22} = 0$. Therefore, the Bayes estimate of μ under SE loss function becomes

$$\hat{\mu}_{SE} \approx \hat{\mu} + \tau_{22}\eta_2 + \tau_{12}\eta_1 + \frac{1}{2} (L_{111}\tau_{11}\tau_{12} + L_{222}\tau_{22}^2 + L_{112}(\tau_{11}\tau_{22} + 2\tau_{12}^2))$$

- Next, considering $h(\theta, \mu) = R(t)$, the approximate Bayes estimate of $R(t)$ against loss function SE is obtained as

$$\hat{R}_{SE}(t) \approx \hat{R}(t; \theta, \mu) + 0.5(R_{11}\tau_{11} + R_{22}\tau_{22} + 2R_{12}\tau_{12}) + \eta_1 V_1 + \eta_2 V_2 + 0.5 (L_{111}\tau_{11}V_1 + L_{222}\tau_{22}V_2 + L_{112}(2\tau_{12}V_1 + \tau_{11}V_2)),$$

Where

$$V_1 = R_1\tau_{11} + R_2\tau_{12},$$

$$V_2 = R_1\tau_{12} + R_2\tau_{22}$$

And

$$R_1 = \frac{\partial R(t, \theta, \mu)}{\partial \mu} = \frac{(1 + \theta(1 + t))}{e^{t\theta}} \frac{\theta^2(1 + \mu)e^{\theta\mu}}{(1 + \theta(1 + \mu))^2}$$

$$R_2 = \frac{\partial R(t, \theta, \mu)}{\partial \theta} = \frac{\theta(\mu^2 + 2\mu - t^2 - 2t) + \theta^2(\mu + \mu^2 + \mu^2t - t - t^2 - \mu t^2)e^{-\theta(t-\mu)}}{(1 + \theta(1 + \mu))^2}$$

$$R_{11} = \frac{\partial^2 R(t, \theta, \mu)}{\partial \mu^2} = \frac{1 + \theta(1 + t)}{e^{t\theta}} \times \frac{(1 + \theta(1 + \mu))^2(\theta^2 e^{\theta\mu} + e^{\theta\mu} \theta^3(1 + \mu)) - 2\theta^3(1 + \mu)e^{\theta\mu}(1 + \theta(1 + \mu))}{(1 + \theta(1 + \mu))^4}$$

$$\begin{aligned} R_{22} &= \frac{\partial^2 R(t, \theta, \mu)}{\partial \theta^2} \\ &= \frac{\theta(\mu^2 + 2\mu - t^2 - 2t) + \theta^2(\mu + \mu^2 + \mu^2t - t - t^2 - \mu t^2)e^{-\theta(t-\mu)}(\mu - t)(1 + \theta(1 + \mu))^2}{(1 + \theta(1 + \mu))^4} \\ &\quad + \frac{(1 + \theta(1 + \mu))^2 e^{-\theta(t-\mu)} \theta(-2t - 2t^2 - 2t^2\mu + 2\mu + 2\mu^2 + 2\mu^2t) + (-2t - t^2 + 2\mu + \mu^2)}{(1 + \theta(1 + \mu))^4} \\ &\quad - \frac{2(\theta(\mu^2 + 2\mu - t^2 - 2t) + \theta^2(\mu + \mu^2 + \mu^2t - t - t^2 - \mu t^2))e^{-\theta(t-\mu)}(1 + \mu)(1 + \theta(1 + \mu))}{(1 + \theta(1 + \mu))^4} \end{aligned}$$

$$\begin{aligned} R_{12} &= \frac{\partial^2 R(t, \theta, \mu)}{\partial \theta \partial \mu} \\ &= \frac{(1 + \theta(1 + t))}{e^{t\theta}} \times \frac{((1 + \theta(1 + \mu))^2(2\theta e^{\theta\mu}(1 + \mu) + \theta^2 e^{\theta\mu} \mu(1 + \mu)) - \theta^2(1 + \mu)e^{\theta\mu} 2((1 + \theta(1 + \mu))(1 + \mu)))}{(1 + \theta(1 + \mu))^4} + \frac{(\theta^2 e^{\theta\mu}(1 + \mu))}{(1 + \theta(1 + \mu))^2} \frac{((1 + t)e^{t\theta} + (1 + \theta(1 + t))te^{t\theta})}{(e^{t\theta})^2} \end{aligned}$$

❖ Bayes Estimates Using Linex(LE) Loss Function:

- The approximate Bayes estimate of θ under the LE loss function is obtained by choosing $h(\theta, \mu) = e^{-\nu\theta}$.

$$\hat{\theta}_{LE} \approx -\frac{1}{\nu} \log(e^{-\nu\hat{\theta}} + 0.5h_{11}\tau_{11} + h_1(\tau_{11}\eta_1 + \tau_{21}\eta_2) + 0.5h_1(L_{111}\tau_{11}^2 + L_{222}\tau_{22}\tau_{12} + 3L_{112}\tau_{12}\tau_{11})),$$

$$\text{where } h_1 = -\nu e^{-\nu\hat{\theta}} \text{ and } h_{11} = \nu^2 e^{-\nu\hat{\theta}}.$$

- Similarly, choosing $h(\theta, \mu) = e^{-\nu\mu}$,

$$\hat{\mu}_{LE} \approx -\frac{1}{\nu} \log(e^{-\nu\hat{\mu}} + 0.5h_{22}\tau_{22} + h_2(\tau_{22}\eta_2 + \tau_{12}\eta_1) + 0.5h_2(L_{111}\tau_{11}\tau_{12} + L_{222}\tau_{22}^2 + L_{112}(\tau_{11}\tau_{22} + 2\tau_{12}^2))),$$

Where $h_2 = -\nu e^{-\nu\hat{\mu}}$ and $h_{22} = \nu^2 e^{-\nu\hat{\mu}}$.

- To evaluate the estimate of $R(t)$ under linex (LE) loss, Let $h(\theta, \mu) = e^{-\nu R(t)}$, and obtain $h_1 = -\nu e^{-\nu\hat{R}(t;\theta,\mu)} R_1$, $h_2 = -\nu e^{-\nu\hat{R}(t;\theta,\mu)} R_2$,

$$h_{11} = \nu e^{-\nu\hat{R}(t;\theta,\mu)} (\nu R_1^2 - R_{11}), h_{22} = \nu e^{-\nu\hat{R}(t;\theta,\mu)} (\nu R_2^2 - R_{22}).$$

$$h_{12} = \nu e^{-\nu\hat{R}(t;\theta,\mu)} (\nu R_1 R_2 - R_{12}).$$

Then

$$\hat{R}_{LE}(t; \theta, \mu) \approx -\frac{1}{\nu} \log(e^{-\nu\hat{R}(t;\theta,\mu)} + 0.5(h_{11}\tau_{11} + h_{22}\tau_{22} + 2h_{12}\tau_{12}) + V_1\eta_1 + V_2\eta_2 + 0.5(L_{111}\tau_{11}V_1 + L_{222}\tau_{22}V_2 + L_{112}(2\tau_{12}V_1 + \tau_{11}V_2))).$$

❖ Bayes Estimates Using General Entropy (GE) Loss Function:

❖ Under GE loss function, the approximate Bayes estimate of θ is obtained by choosing $h(\theta, \mu) = \theta^{-\omega}$, Thus, we have

$$\hat{\theta}_{GE} \approx [\hat{\theta}^{-\omega} + 0.5h_{11}\tau_{11} + h_1(\tau_{11}\eta_1 + \tau_{12}\eta_2) + 0.5h_1 (L_{111}\tau_{11}^2 + L_{222}\tau_{12}\tau_{22} + 3L_{112}\tau_{12}\tau_{11})]^{-\frac{1}{\omega}},$$

Where $h_1 = -\omega\theta^{-\omega-1}$ and $h_{11} = \omega(\omega + 1)\theta^{-\omega-2}$.

• Similarly , for the parameter μ , we use $h(\theta, \mu) = \mu^{-\omega}$ and obtain

$$\hat{\mu}_{GE} \approx [\hat{\mu}^{-\omega} + 0.5h_{22}\tau_{22} + h_2(\tau_{22}\eta_2 + \tau_{12}\eta_1) + 0.5h_2 (L_{111}\tau_{11}\tau_{12} + L_{222}\tau_{22}^2 + L_{112}(\tau_{11}\tau_{22} + \tau_{12}^2))]^{-\frac{1}{\omega}},$$

Where $h_2 = -\omega\mu^{-\omega-1}$ and $h_{22} = \omega(\omega + 1)\mu^{-\omega-2}$.

• Finally, the reliability $R(t)$ can be estimated by considering $h(\theta, \mu) = R(t)^{-\omega}$. then , we have

$$\hat{R}_{GE}(t; \theta, \mu) \approx [\hat{R}^{-\omega}(t; \theta, \mu) + 0.5(h_{11}\tau_{11} + h_{22}\tau_{22} + 2h_{12}\tau_{12}) + V_1\eta_1 + V_2\eta_2 + 0.5 (L_{111}\tau_{11}V_1 + L_{222}\tau_{22}V_2 + L_{112}(2\tau_{12}V_1 + \tau_{11}V_2))]^{-\frac{1}{\omega}}$$

Where

$$\begin{aligned} h_1 &= -\omega R(t)^{-\omega-1} R_1, \\ h_2 &= -\omega R(t)^{-\omega-1} R_2, \\ h_{11} &= \omega(\omega + 1)R(t)^{-\omega-2} R_1^2 - \omega R(t)^{-\omega-1} R_{11}, \\ h_{22} &= \omega(\omega + 1)R(t)^{-\omega-2} R_2^2 - \omega R(t)^{-\omega-1} R_{22}, \\ h_{12} &= \omega(\omega + 1)R(t)^{-\omega-2} R_1 R_2 - \omega R(t)^{-\omega-1} R_{12}. \end{aligned}$$

Maximum Likelihood Estimation(MLE) of Parameters:

The likelihood function for a random sample $X_1, X_2, \dots, X_n$ which is taken from $SL(\mu, \theta)$ distribution is given by:

$$L(X, \mu, \theta) = \frac{\theta^{2n}}{(1 + \theta(1 + \mu))^n} \prod_{i=1}^n (1 + x_i) e^{-\theta \sum_{i=1}^n (x_i - \mu)}$$

- Then the log-likelihood function is given by

$$\log(L(X, \mu, \theta)) = (2n \log \theta) - n \log(1 + \theta(1 + \mu)) + \sum_{i=1}^n \log(1 + X_i) - \theta \sum_{i=1}^n (X_i - \mu).$$

- The maximum likelihood estimator (MLE) of μ is:

$$\hat{\mu}_{MLE} = \min_i X_i = X_{(1)}.$$

- Differentiating the log-likelihood with respect to θ , we get the following equation:

$$\frac{2n}{\theta} - \frac{n(1 + \hat{\mu}_{MLE})}{1 + \theta(1 + \hat{\mu}_{MLE})} - \sum_{i=1}^n (X_i - \hat{\mu}_{MLE}) = 0.$$

Simulation:

❖ Algorithm for Data Simulation:

- Select values of θ and μ
- Calculate weight $w = \frac{(1 + \mu)\theta}{1 + \theta(1 + \mu)}$
- Generate U from U(0,1)
- If $U < w$, generate a sample from $f_{SG}(x; 1, \theta, \mu)$ else from $f_{SG}(x; 2, \theta, \mu)$ using `rgamma3(1,shape,scale,thres)`

➤ Then, we have used two sets of parameter values

- Set 1: $(\theta, \mu) = (2, 1)$
- Set 2: $(\theta, \mu) = (1.5, 1.5)$

➤ Choices for v and ω :

- v : -0.5, 1, 1.5
- ω : -0.5, 1, 1.5

➤ To evaluate the Bayes estimates, we take 2 of hyper-parameter values -

- $a1=a2=b1=b2=2$

○ Then, approximate Bayes estimates of the parameter θ, μ and $R(t)$ are computed by using Lindley method . Note that all the ML and Bayes estimators of $R(t)$ are evaluated at $t=1$. All the calculations are conducted using R programming.

• Simulation Results Under Set 1: $(\theta, \mu) = (2, 1)$:

Table 1: $\theta=2,\mu=1$

n		MLE	L_{SE}	$\nu = -0.5$	L_{LE}		$\omega = -0.5$	L_{GE}	
					$\nu=1$	$\nu=1.5$		$\omega =1$	$\omega=1.5$
15	$\hat{\theta}$	1.9209	1.9695	1.9715	1.9652	1.9628	1.9684	1.9650	1.9638
	$\hat{\mu}$	1.0169	1.1419	1.1321	1.1649	1.1791	1.1523	1.1927	1.2107
	$R(\hat{t})$	1.0261	0.8464	0.8609	0.6288	0.6426	0.9495	0.7648	0.7921
20	$\hat{\theta}$	2.0001	2.0364	2.0381	2.0329	2.0310	2.0356	2.0329	2.0319
	$\hat{\mu}$	1.0169	1.1099	1.1037	1.1242	1.1324	1.1166	1.1405	1.1503
	$R(\hat{t})$	1.0274	0.8880	0.9010	0.7039	0.7101	0.9697	0.8110	0.8327
30	$\hat{\theta}$	2.2666	2.2902	2.2915	2.2877	2.2863	2.2897	2.2880	2.2874
	$\hat{\mu}$	1.0121	1.0732	1.0701	1.0798	1.0834	1.0763	1.0870	1.0911
	$R(\hat{t})$	1.0227	0.9203	0.9321	0.7720	0.7724	0.9836	0.8520	0.8680
50	$\hat{\theta}$	2.0755	2.0896	2.0903	2.0879	2.0873	2.0892	2.0880	2.0877
	$\hat{\mu}$	1.0004	1.0373	1.0353	1.0413	1.0434	1.0392	1.0456	1.0480
	$R(\hat{t})$	1.0006	0.9441	0.9508	0.8618	0.8592	0.9749	0.9028	0.9107
70	$\hat{\theta}$	1.9804	1.9905	1.9910	1.9893	1.9888	1.9902	1.9894	1.9891
	$\hat{\mu}$	1.0003	1.0269	1.0254	1.0298	1.0314	1.0284	1.0330	1.0346
	$R(\hat{t})$	1.0005	0.9617	0.9663	0.9041	0.9012	0.9822	0.9317	0.9371

- From table 1 it can be seen that-
- The Bayes estimates of θ is almost similar under different loss function.
 - Similar feature can be seen for Bayes estimate of μ .
 - The absolute bias in case of MLE of μ is getting reduced as sample size increases.

- Bayes estimates of $R(1)$ is getting close to 1 as the sample size increased.
- The MLE always overestimates the $R(1)$ value where the Bayes estimates under all loss functions, underestimates $R(1)$.
- The smallest estimate of $R(1)$ is 0.7921 which is achieved when generalized entropy loss function was used with $\omega = 1.5$.
- Larger sample sizes result in more stable and accurate estimates across all methods.
- The L_{GE} estimates (with higher ω values) tend to give slightly higher estimates compared to others, especially for smaller sample sizes.
- For larger sample sizes, all methods converge towards the true values, but LGE with $\omega=1.5$ shows slightly higher variability.

• Simulation Results Under Set 2: $(\theta, \mu) = (1.5, 1.5)$:

Table 2: $\theta=1.5, \mu=1.5$

n		MLE	L_{SE}	$\nu=-0.5$	L_{LE}		L_{GE}		
					$\nu=1$	$\nu= 1.5$	$\omega=-0.5$	$\omega= 1$	$\omega=1.5$
15	$\hat{\theta}$	1.4518	1.5289	1.5290	1.5285	1.5281	1.5288	1.5282	1.5279
	$\hat{\mu}$	1.5225	1.6717	1.6559	1.7108	1.7362	1.6831	1.7249	1.7425
	$\hat{R}(t)$	1.7879	1.5698	1.4495	0.7040	0.6702	2.0913	1.0655	1.1587
20	$\hat{\theta}$	1.5114	1.5693	1.5697	1.5683	1.5678	1.5690	1.5680	1.5677
	$\hat{\mu}$	1.5225	1.6318	1.6216	1.6558	1.6701	1.6391	1.6641	1.6740
	$\hat{R}(t)$	1.8414	1.6678	1.5709	0.8817	0.8126	2.0921	1.1942	1.2730
30	$\hat{\theta}$	1.7122	1.7506	1.7511	1.7495	1.7489	1.7503	1.7493	1.7489
	$\hat{\mu}$	1.5161	1.5847	1.5798	1.5956	1.6017	1.5881	1.5994	1.6035
	$\hat{R}(t)$	2.0170	1.8764	1.7878	1.1185	0.9928	2.2484	1.3971	1.4591
50	$\hat{\theta}$	1.5684	1.5912	1.5916	1.5905	1.5900	1.5910	1.5903	1.5900
	$\hat{\mu}$	1.5004	1.5432	1.5399	1.5501	1.5538	1.5455	1.5526	1.5551
	$\hat{R}(t)$	1.8426	1.7709	1.7367	1.3433	1.2376	1.9406	1.5126	1.5884
70	$\hat{\theta}$	1.4968	1.5131	1.5134	1.5125	1.5122	1.5129	1.5123	1.5121
	$\hat{\mu}$	1.5004	1.5318	1.5293	1.5369	1.5397	1.5333	1.5387	1.5405
	$\hat{R}(t)$	1.7810	1.7332	1.7128	1.4343	1.3473	1.8424	1.5561	1.5811

➤From table 2 it can be seen that-

- The Bayes estimates of θ is almost similar under different loss function.
- All methods provide relatively accurate estimates of θ and μ ,with slight overestimations.
- MLE and L_{SE} estimators provide close and consistent estimates for θ and μ .

- L_{GE} estimators with higher values of ω and v provide higher estimates for μ and $R(t)$ compared to other methods. LSE seems to offer reliable estimates across all sample sizes, particularly for θ . No single estimator is universally best for all parameters and sample sizes.
- For smaller sample sizes, L_{SE} and MLE provide more reliable estimates.
- For larger sample sizes, all methods converge towards the true values, but L_{GE} with $\omega=1$ or 1.5 shows slightly higher variability.

Data analysis:

- The proposed distribution is fitted for a data set available in Duffy et al.(1993).
- The data consists of measurements on strength of the sintered silicon nitride.
- The data is-
613.9,623.4,639.3,642.1,653.8,662.4,669.5,672.8,681.3,682.0,699.0,714.5,717.4,725.5,741.6,744.9,751.0,761.7,763.9,774.2,791.6,795.2,829.8,838.4,856.4,868.3,882.9
- The reliability $R(t)$ at $t=1, 1.5$ are computed.

➤ Choices for ν and ω :

- ν : -0.5, 1, 1.5
- ω : -0.5, 1, 1.5

➤ To evaluate the Bayes estimates, we take 2 of hyper-parameter values -

- $a_1=a_2=b_1=b_2=2$

- Data analysis results:

Table 3

		θ	μ	R(t)	
				$t = 1$	$t = 1.5$
MLE		1.0897	1.1391	1.1106	0.7544
L_{SE}		1.1222	1.2332	1.1356	0.7706
L_{LE}	$\nu=-0.5$	1.1229	1.2193	1.1398	0.7837
	$\nu= 1$	1.1206	1.2653	0.9859	0.6542
	$\nu=1.5$	1.1197	1.2846	0.9746	0.6548
L_{GE}	$\omega =-0.5$	1.1214	1.2464	1.1854	0.8048
	$\omega =1$	1.1191	1.2948	1.0557	0.7201
	$\omega=1.5$	1.1184	1.3158	1.071	0.7316

- From table 3 it can be seen that-
- Maximum likelihood estimates of θ , μ are almost similar with the estimates under loss functions.
- When $t=1$,all values of $R(t)$ are approximately same except L_{LE} with $\nu=1,\nu=1.5$.
- Similarly, for $t=1.5$, similar observation can be seen under Linex loss function with $\nu=1$ and $\nu = 1.5$.
- The estimates of $R(t)$ is small pretty small compared to maximum likelihood estimates.

Discussion:

- In this work, efforts were made to get Bayesian inference on a newly derived distribution based on different loss functions.
- The simulation studies showed satisfactory results in term of biases and sd values.
- From the data analysis, the estimates are found to be reasonable one.
- Comparing the approximate method with MCMC under the present set up will also be interesting which will be pursued in future.